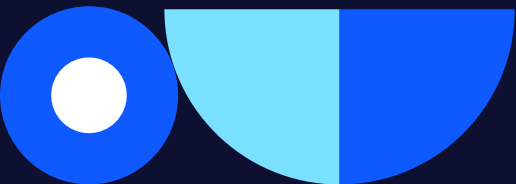




Quantum Foundations

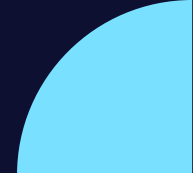
From bits to qubits: the ideas behind quantum computing

[Person 1] · Big Data seminar
Quantum Computing · Part 1 of 3





Roadmap



01

Why quantum

Two limits of ordinary computers

02

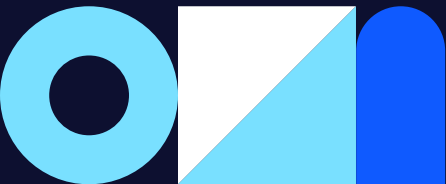
One qubit

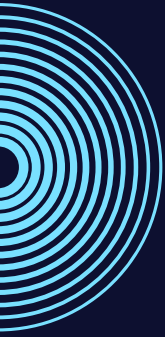
Mixing, reading and interference

03

Many qubits

Entanglement and no-cloning

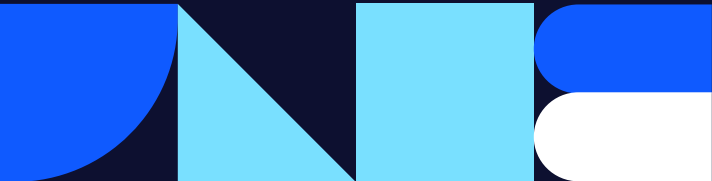




01

Why quantum?

Two reasons ordinary computers are running out
of room



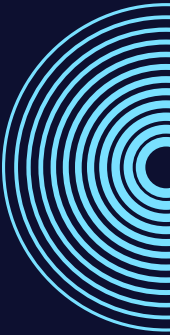
Why quantum computing?

Limit 1: chips

Transistors are now only a few dozen atoms wide. Any smaller and quantum effects break the circuit.

Limit 2: nature

To imitate a quantum system, every extra particle doubles the numbers a normal computer must store.





Rice on a chessboard



50 particles

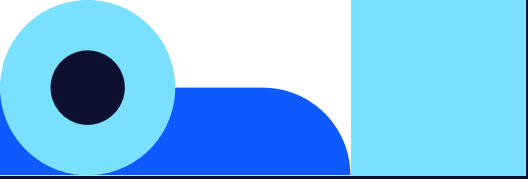
about a million billion numbers

300 particles

more numbers than atoms in the universe

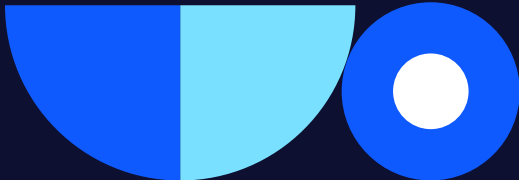
Each square doubles the one before. A quantum system does the same with every extra particle.

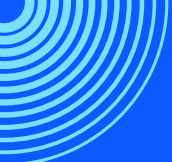




“...if you want to make a simulation of
nature, you'd better make it quantum
mechanical.”

—Richard Feynman (lecture 1981, published 1982)

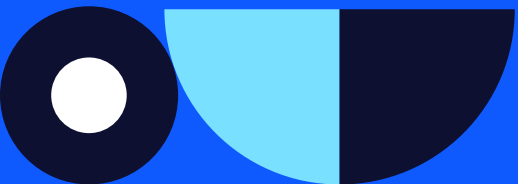




The takeaway

**A specialist for certain
problems, not a faster laptop.**

Next: how one qubit works.





02 One qubit

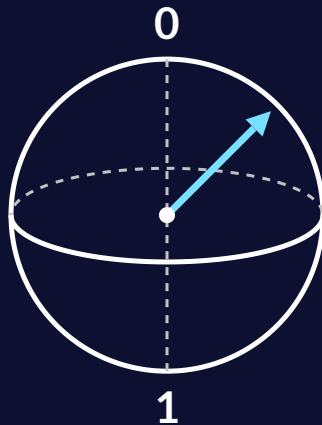
What a qubit is, and what happens when you read it



Bit vs qubit



**Bit: a switch,
only 0 or 1**



Qubit: an arrow on a sphere

Up = 0, down = 1, in between = a mix

Each part of the mix has a **weight**.
The bigger the weight, the more
likely you read that value.

Physicists write the two basic states as $|0\rangle$ and $|1\rangle$. The brackets are just name tags.

A bit is a switch. A qubit is an arrow that can point anywhere on a sphere.





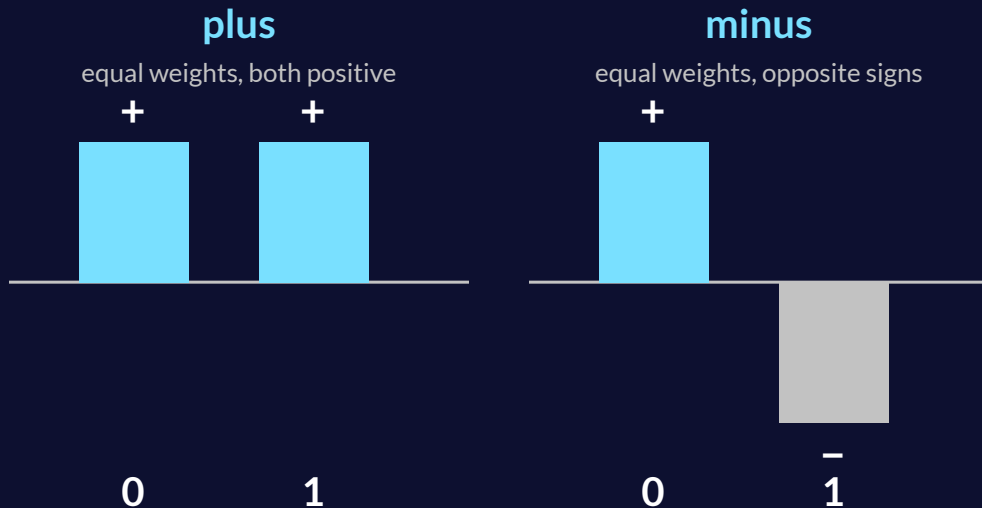
Bit and qubit, side by side

	Bit	Qubit
What it holds	0 or 1	0, 1 or a mix of both
Reading it	gives the stored value	gives 0 or 1 at random
After reading	unchanged	changed for good
Copying	easy	impossible





0 and 1 at the same time



Same size, same 50/50 when read. Only the sign differs.

Superposition: a qubit holding 0 and 1 at once, each with its own weight.

Weights can be **positive or negative**. Ordinary chances can't be. This is what makes quantum different.

Plus and minus both read as 50% 0 and 50% 1, yet they are different states.



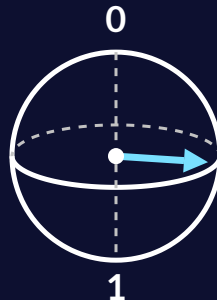
Two common myths

✗ Hidden coin



Already heads or tails,
we just can't see it.
Not how a qubit works

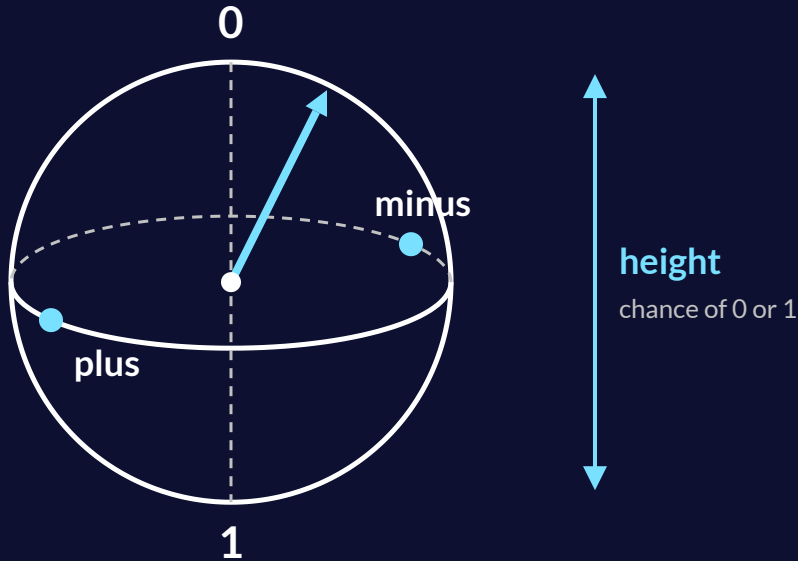
✓ Qubit in superposition



Genuinely undecided until read,
and the arrow also has a direction.
How a qubit really behaves

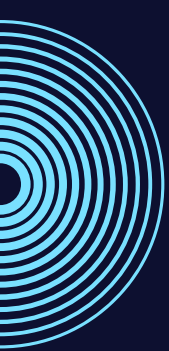
Also not “trying every answer at once”: reading gives back only one result.

Picturing a qubit



- North pole = 0, south pole = 1, equator = the 50/50 mixes
- **Height** sets the chance of reading 0 or 1
- **Direction around** is the **phase**: a plain reading can't see it, but it changes what happens next
- Plus and minus: same height, opposite direction

The Bloch sphere. Quantum gates simply rotate this arrow (Part 2). It works for one qubit only.

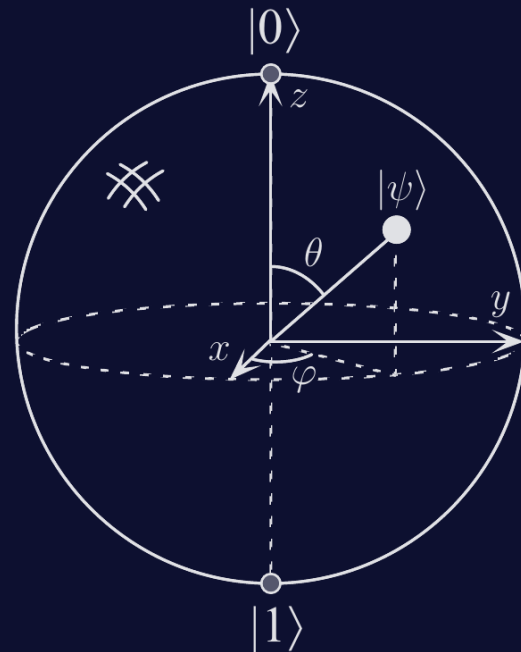


The math of one qubit

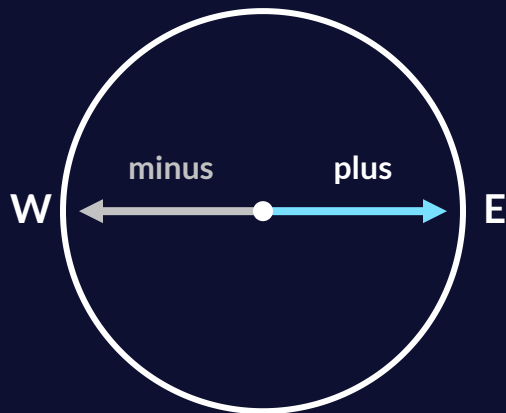
A qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. Reading gives 0 with chance $|\alpha|^2$ and 1 with chance $|\beta|^2$.

plus = $(|0\rangle + |1\rangle)/\sqrt{2}$, minus = $(|0\rangle - |1\rangle)/\sqrt{2}$. On the sphere: $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi} \sin(\theta/2)|1\rangle$, with θ the height and φ the phase.

Source: Nielsen & Chuang (2010), Fig. 1.3.



The hidden direction



The equator seen from above

Ask “0 or 1?”

plus $\rightarrow$ 50/50 $\cdot$ minus $\rightarrow$ 50/50

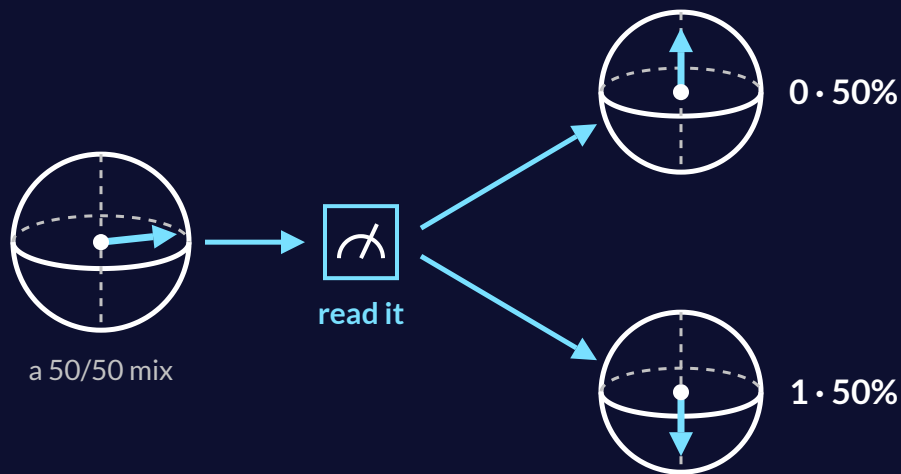
They look identical.

Ask “east or west?”

plus $\rightarrow$ always E $\cdot$ minus $\rightarrow$ always W

The phase is real information.

Reading: a one-way door

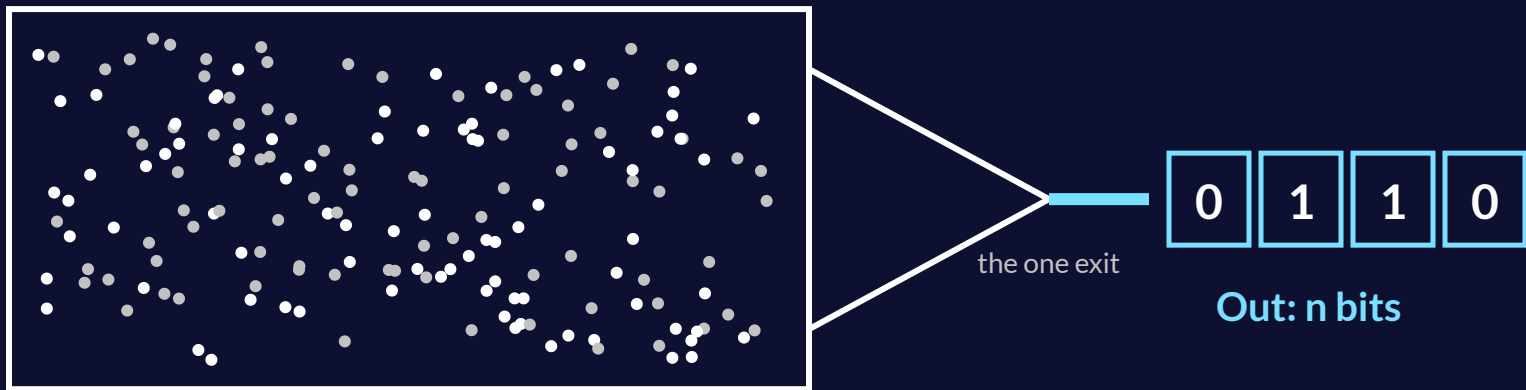


Random: even if you know the qubit exactly, you only know the odds, never the single result.

Collapse: after reading, the qubit becomes the answer. The mix is gone for good.

After reading, the arrow snaps to a pole and everything else is lost.

A huge store, a tiny door

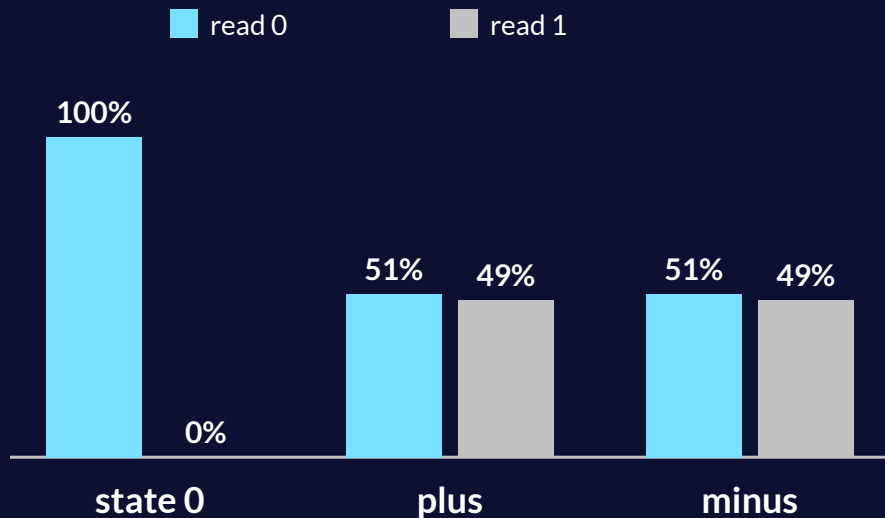


Inside: 2^n weights at once

Out: n bits

Reading gives back only one bit per qubit: a huge store with a tiny door.

What our run showed



51% / 49%

for both plus and minus: they look identical

100%

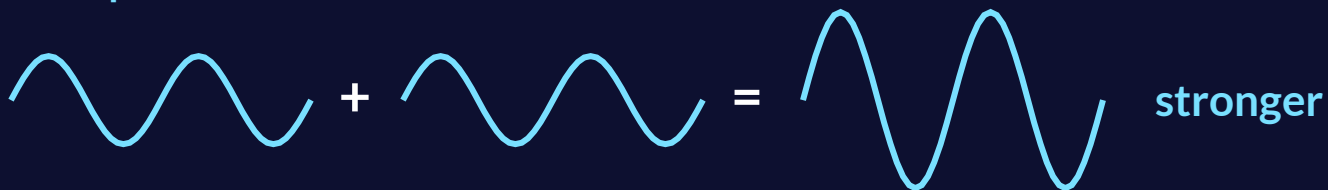
zeros for state 0, every time

That is why quantum programs are run thousands of times.

Our run: 1,024 runs per state, on an ideal noise-free simulator.

Weights act like waves

In step



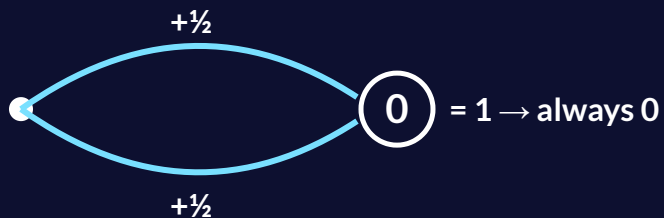
Out of step



Weights behave like waves: in step they add up, out of step they cancel.

Weights can cancel

Plus, then H



Minus, then H



Add one extra step (a gate called H) before reading. Each result can then be reached along two paths.

100%

Our run, 1,024 runs: plus always gives 0, minus always gives 1

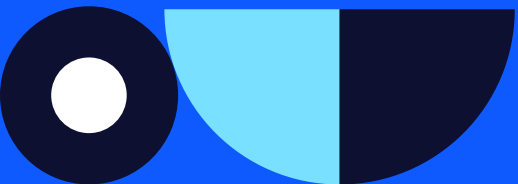
Equal signs add up, opposite signs cancel.

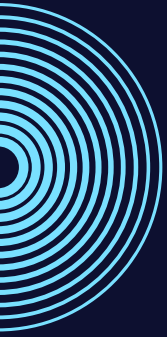


The key idea

**Ordinary chances only add up.
Quantum weights can cancel.**

Algorithms boost the right answer and cancel the wrong ones.





03

Many qubits

How qubits combine, and what they can never do





From one qubit to many

1 qubit

0	1
---	---

 = 2

2 qubits

00	01	10	11
----	----	----	----

 = 4

3 qubits

000	001	010	011	100	101	110	111
-----	-----	-----	-----	-----	-----	-----	-----

 = 8

n qubits

2^n combinations,
each with its own weight

Independent qubits: each can be described on its own. Easy for a normal computer.

Entangled qubits: only the whole group can be described. All 2^n weights are needed.

This is the doubling from the start. Entanglement is a big reason quantum computers are so hard to imitate.



Two qubits, one story

The simplest entangled pair is called a “Bell pair” (Part 2 shows how to make one).



Read one qubit and the other always agrees, however far apart.

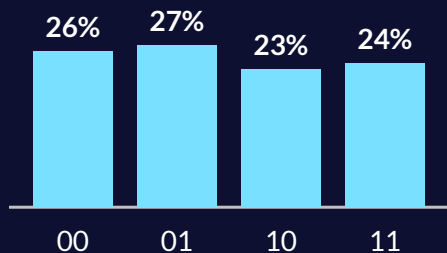
Each qubit alone is pure chance. The information lives in the link between them.

No faster-than-light messages: each side sees only random bits until they compare notes.

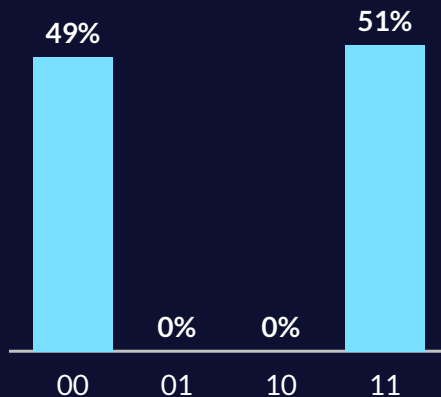


Independent vs entangled

Two independent qubits



Entangled (Bell) pair



$\approx 25\%$

for each result, with independent qubits

49% / 51%

only 00 and 11, with the entangled pair

01 or 10: never.

Our run: 1,024 runs per experiment, on an ideal noise-free simulator.



Two gloves in two boxes?

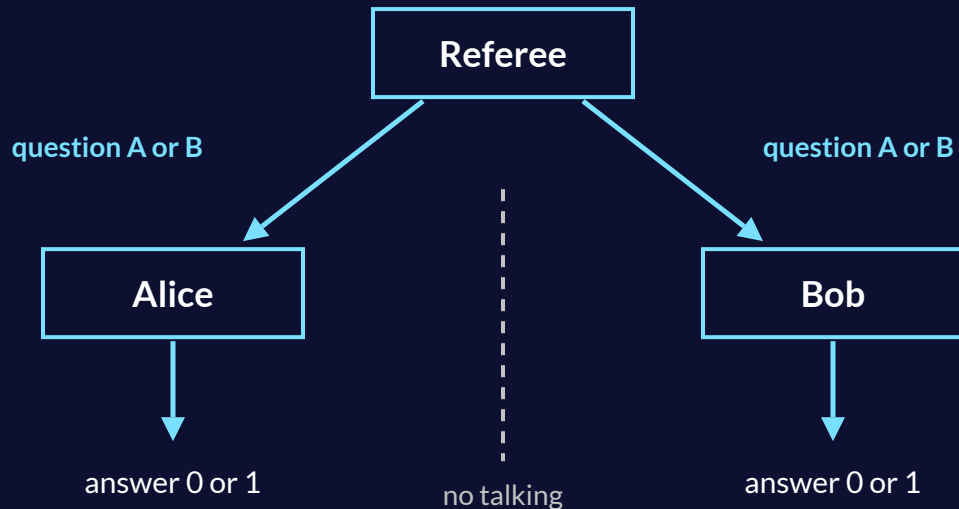
Einstein (1935) asked: does the pair simply carry answers fixed in advance, like two gloves?



Open one box and you instantly know the other.

Nothing strange: the answers were fixed from the start.

Bell's game



Alice and Bob each get question A or B at random and answer 0 or 1, with no talking.

They win if the answers match, unless both got B: then the answers must differ.

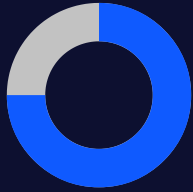
Bell's game, also called the CHSH game.

Bell test: the setup



Source: Nielsen & Chuang (2010), Fig. 2.4. Alice and Bob each choose one of two measurements.

Stronger than ordinary links



75%

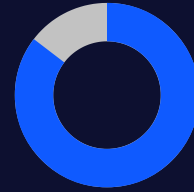
Fixed plan

Best plan agreed in advance

85.1%

Entangled

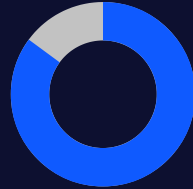
Entangled qubits, in our
simulation



85.4%

Theory

The most quantum physics
allows

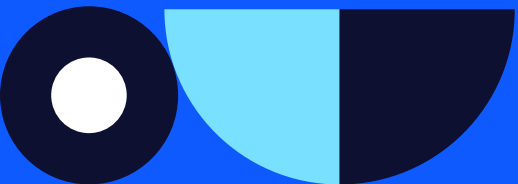




Bell test result

**Entanglement is not answers
fixed in advance. It is a
genuinely new resource.**

Confirmed by real experiments: Nobel Prize 2022 (Aspect, Clauser, Zeilinger).

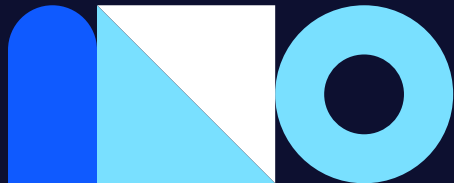




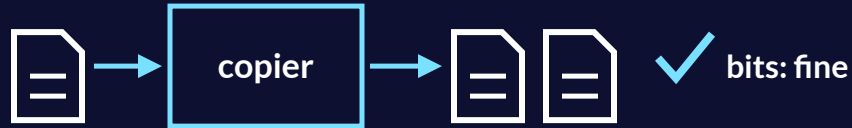
Math: many qubits, Bell

The numbers behind the last few slides:

- n qubits: 2^n complex weights; independent qubits need only 2 per qubit
- Bell pair: $(|00\rangle + |11\rangle)/\sqrt{2}$
- Bell's game: best fixed plan 75%; entangled pair $\cos^2(\pi/8) \approx 85.4\%$, the most quantum physics allows



You cannot copy a qubit



Rule: no machine can copy an unknown qubit.

A machine that copies 0 and 1 correctly turns a mix into a linked pair, not two copies.



Bits are fine, which is why ordinary data can be copied freely.

Reading first doesn't help: one reading gives one bit and destroys the mix.

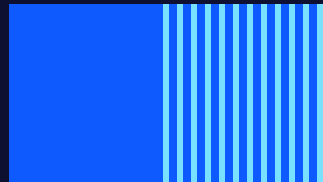
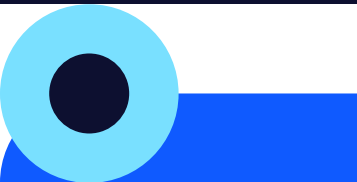


Math: why copying fails

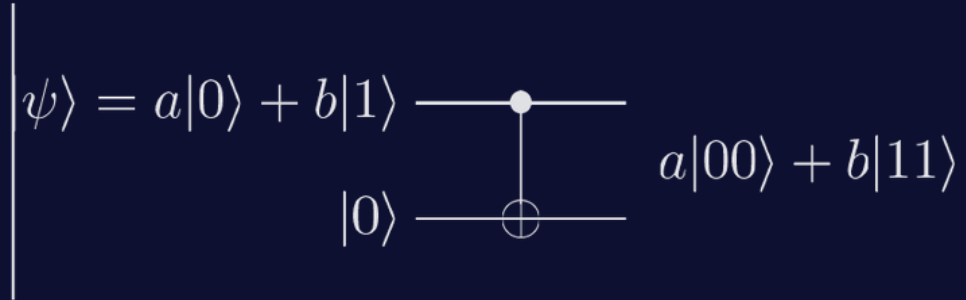
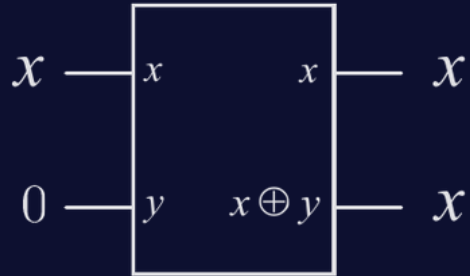
A copier that maps $|0\rangle|0\rangle \rightarrow |0\rangle|0\rangle$ and $|1\rangle|0\rangle \rightarrow |1\rangle|1\rangle$ must, by linearity, send $\alpha|0\rangle + \beta|1\rangle$ to $\alpha|00\rangle + \beta|11\rangle$: an entangled pair (circuit on the next slide).

A true copy would be $(\alpha|0\rangle + \beta|1\rangle)(\alpha|0\rangle + \beta|1\rangle)$, a different state. So no such copier can exist.

Source: Nielsen & Chuang (2010), Box 12.1; Wootters & Zurek (1982).

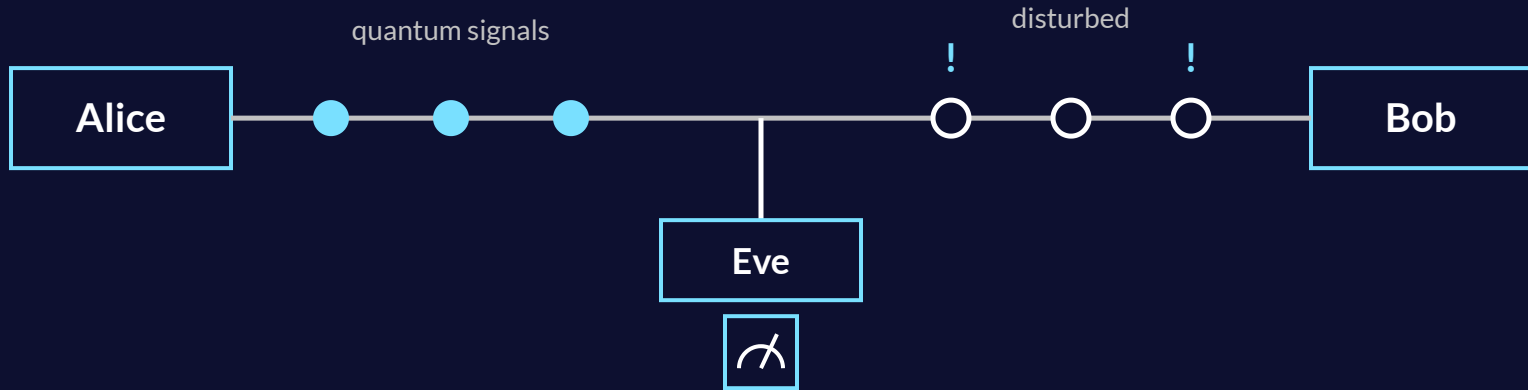


Copying a bit, then a qubit



Source: Nielsen & Chuang (2010), Fig. 1.11. The bit-copying circuit turns $a|0\rangle + b|1\rangle$ into $a|00\rangle + b|11\rangle$.

Good news, bad news



Good news: Eve cannot copy the signals, so she must read them, and reading leaves errors.

Alice and Bob compare a sample, see the errors and know someone listened.

Bad news: no backup, replication or caching of quantum data.



Three quantum resources

01

Superposition

Opens up 2^n possibilities

02

Entanglement

Links qubits into one system

03

Interference

Pushes the odds toward the right answer

Two rules to remember: reading gives only one bit per qubit · qubits cannot be copied.





Thank you!

Next, the gates and circuits that put these resources to work.

Part 2 • Gates, Circuits & Algorithms

